Problem-Solution Fit

Date	01 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS]

- Farmers
- Agricultural officers
- Agricultural researchers
- Students studying agriculture
- Government agriculture departments
- NGOs supporting farmers
- Agri-tech startups

6. CUSTOMER LIMITATIONS
EG. BUDGET, DEVICES **[CL]**

- Limited technical knowledge.
- Limited internet access.
- Budget constraints.
- Lack of soil testing facilities.
- Limited awareness of AI-based farming tools.

5. AVAILABLE SOLUTIONS
PROS & CONS **[AS]**

Pros

- Government agriculture portals provide recommendations.
- Agricultural experts offer guidance.
- Existing crop recommendation tools are available.

Cons

- Recommendations may not be location-specific.
- Manual consultation takes time.
- Limited accessibility in remote areas.
- Some tools require internet connectivity.

• Difficulty in selecting the most suitable crop. • Lack of knowledge about soil nutrient requirements. • Unpredictable weather affects crop planning. • Low crop yield due to incorrect crop selection. • Manual decision-making is time-consuming.	**9. PROBLEM ROOT / CAUSE [RC]** • Lack of accurate crop recommendation systems. • Variation in soil nutrients across regions. • Dependence on traditional farming methods. • Limited access to real-time agricultural data. • Insufficient awareness of AI technologies.	**7. BEHAVIOR** + ITS INTENSITY **[BE]** • Frequently searches for crop recommendations. • Relies on traditional farming practices. • Uses weather forecasts before sowing. • Consults local agricultural experts. • Checks soil conditions before cultivation.
3. TRIGGERS TO ACT [TR] • Need to improve crop productivity. • Seasonal crop planning. • Government agricultural initiatives. • Demand for data-driven farming. • Better utilization of soil and water resources.	**10. YOUR SOLUTION [SL]** Develop an **AI-Based Crop Recommendation System (OptiCrop)** using **Flask** and **Machine Learning** that predicts the most suitable crop based on **Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall**. The system provides accurate, fast, and user-friendly crop recommendations to help farmers improve productivity and make informed agricultural decisions.	**8. CHANNELS OF BEHAVIOR [CH]** ONLINE • Web application • GitHub • Research websites
Before • Confused • Uncertain about crop selection • Worried about crop failure • Fear of financial loss		OFFLINE • Government offices • Research organizations • Libraries

After

- Confident
- Better crop planning
- Improved productivity
- Increased satisfaction